Diploma in UX Design

UX Research

Summary Notes

Contents

3 Lesson objectives

3 Introduction

3 Introduction to UX

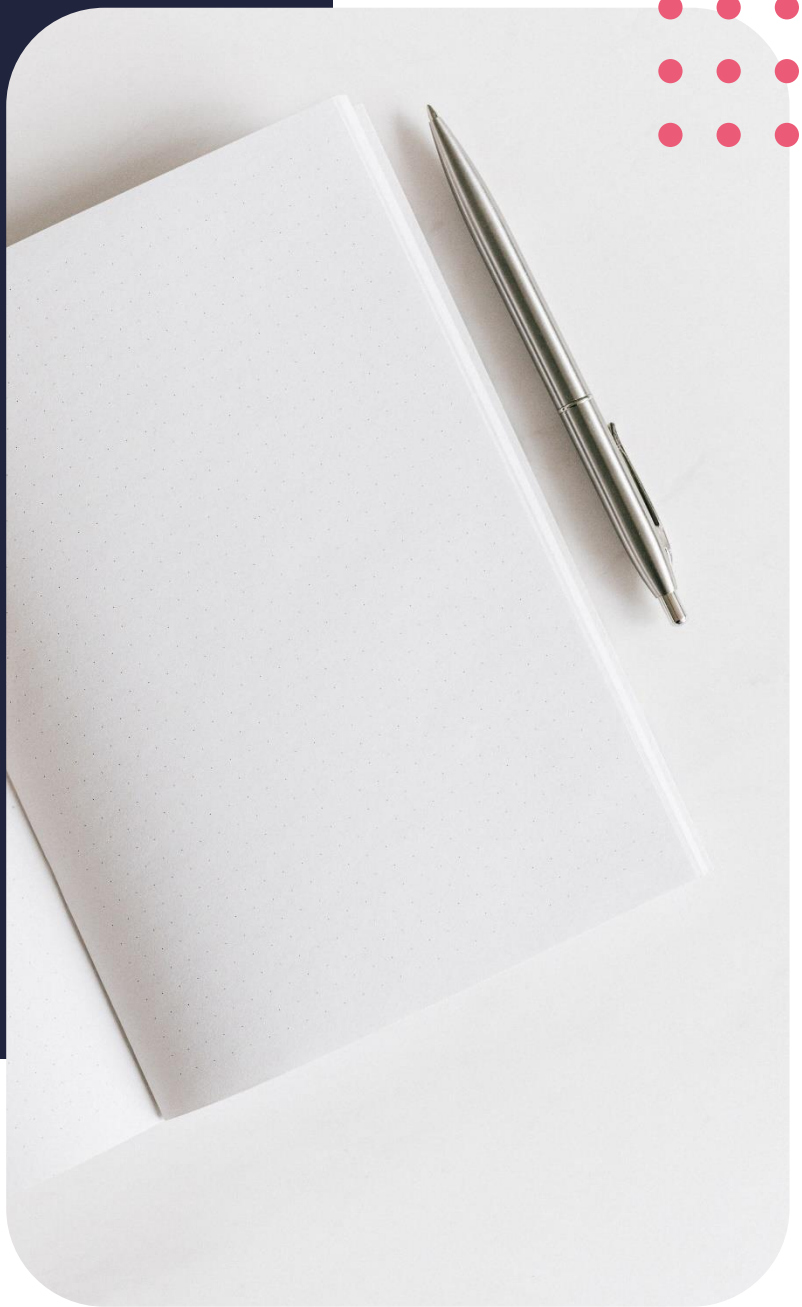
5 Research participant selection and recruitment

7 User testing

8 Lesson challenge

10 Conclusion

10 References



Lesson objectives

By the end of this lesson, you should be able to:

- Understand to UX research
- Investigate research participant selection
- Understand user-testing

Introduction

UX research aims to gather more information and insights about your target users and your competitors so that you can get ideas and make improvements to serve their needs better or even reduce their pain points to a larger degree. We will investigate what exactly UX research entails, why it is important, types of research and how to get people to take part in your research. Lastly, we will learn about the most common UX research technique namely user-testing.

Introduction to UX research

What is UX research?

User Experience (UX) research is any activity undertaken to gather insights into:

1) Who the current, potential, or possible future users of a digital product are.

2) What their needs and goals are, what features and content they value most, and what tasks they must do and how frequently. How a product can be (re)designed to better meet their needs and quicker, easier, better, and cheaper enable them to reach their goals.

3) What designs, configurations and options users find most attractive.

4) How functionality, content and design can be tailored to an individual level to provide a personalised offering according to the needs, preferences, and context of the specific user.

5) What are their pain points in general, and in using your product, and how can your product reduce or eliminate those pain points and make the user experience seamless and pleasant?

6) In what context, under what circumstances, and during what times or events, and with what types of devices are or will the product be used, and how it can be designed to be easier to use under those circumstances.

7) What is your target market's technical, domain or subject knowledge and level of experience? Or are your target users' experience so that the product and its functionality and content will not be too simplistic or complicated for them.

8) What are similar products in your market or in other markets from where you can get ideas, their functionalities and content strategy? In terms of which user needs, functionality and content are you behind, on par or better than them? Are you better than them to sustain a strong competitive advantage, or how are you different to them so that you can occupy a particular market niche?

9) What are the latest technologies, trends, and opportunities out there that you can take advantage of to meet the needs of your target users better or to cut costs?

Why is UX research important?

UX research is a very important skill for any UX designer because in practice you will have to do the research yourself, especially if you work on a small project or for a small company. But even if the research will be conducted by specialist researchers, you may be tasked to oversee them, assist them or check the quality of the research process they adopted, and most importantly, you have to implement their insights into your designs. It is highly recommended for UX designers to frequently engage in research themselves because there is no better way to learn and improve your experience and intuition for good design than by directly engaging with actual users. Also, many UX designers progress their careers into UX research.

Insights from UX research are crucial for the success of a digital product because of several reasons:

1) If you know exactly what users want and how to improve your product to be much more useful, enjoyable and painless to use, you will gain market share and increase your revenue.

2) If you know what features and content users' value most, you can save costs by focusing on those features only and develop them very well, not wasting expensive development, marketing and commercialisation costs on other features and content that users will not want to use or pay for;

3) You can easily test assumptions and which design, feature and content options are most effective, thereby resolving different opinions and disputes amongst team members and make rational decisions.

4) If you know what new technologies, trends, and gaps in the market there are, you can grab those opportunities before your competitors do, thereby gaining or maintaining market share and possibly get a first-mover advantage by being the first in the market to capitalise on those.

5) You need to know what your competitors are doing and to continuously benchmark your product against theirs because it is essential to remain superior or provide the best value in your market niche otherwise your users will switch to the better competitor very quickly.

Research types

There are two types of research: Foundational or generative research; and experimental and evaluative research.

Foundational or generative research is traditionally performed at the beginning of the product's life cycle, to learn about who the target users are, their context, needs and pain points or problems they have that you can solve. What product or feature opportunities there are? It can also be used at later stages to learn even more about the users, context and so on, especially in the context of changing competitive and technological environments, new trends and changing user habits and expectations, and to uncover ideas for further product improvement. It is essential to inform medium to longer-term product and business strategy. Foundational or generative research includes interviews, surveys, focus groups, ethnography, and various others that we will discuss in a future lesson.

Experimental and evaluative research is usually conducted at the later stages of development where a prototype or actual product already exists, to uncover shorter-term strategies to improve or optimise the user experience. It could also be conducted during the early stages when competitive or similar products are tested as part of a competitor analysis or benchmarking study, or to get ideas for replication and improvement. Later in this lesson, we will discuss User Testing as an evaluative or experimental UX research technique, and in a future lesson, we will also look at other evaluative techniques such as A/B testing and First Click tests.

Did you know?

There are multiple benefits of user testing and it is staggering to see that only 55% of companies conduct this type of testing. The main reason behind this might the a matter of time constraints but based on the fact that 85% of issues can be picked up during testing, it would make sense to make UX testing a part of any UX design process.

Research participant selection and recruitment

Although some UX research techniques like literature reviews and benchmarking studies do not require the participation of people or actual users, most techniques do. Finding people to participate in your research is referred to as user recruitment. Different UX research techniques require different recruitment methods. This section will describe the recruitment techniques for user testing, which can be adapted to suit the needs of other techniques.

Criteria for selecting research participants

The research participants must be representative of a subset of the target user group. They should be like the personas you design for, assuming that the personas have been accurately defined in terms of relevant characteristics to represent a real target user subgroup. Relevant characteristics are highly dependent on the product, but may include demographics that are relevant to the product, need for the product, buying habits and means, career or roles that entail actual or potential usage of the product, product usage patterns, product involvement and experience level, domain or subject knowledge and interest, similar behaviour patterns, key personality characteristics, customer channels used, platforms or devices used and experience using it, and other relevant information from the persona or customer journey. Although it is important to test all types of users including the less motivated and knowledgeable, people who are highly interested and knowledgeable about a particular domain or area of interest may provide very rich insights because they are eager to test and comment on the functionality and already know a lot about alternatives and how other apps solved similar issues or have greater features.

Finding research participants

There are several approaches you can adopt to find people to participate in your research, including:

a) Colleagues, friends, and family:

b) Recruitment firms and temp agencies: A recruitment firm or temp staff agency can be requested to find participants who meet the identified criteria. Although it makes the research process easier because they can handle this potentially difficult and very time-consuming step, this may be a very expensive option, and if it is an unreliable firm, chances are good that they may not recruit the right type of respondent. Temp agencies may have access to only a limited type of user.

c) Online user research platforms: Platforms such as OpinionAPP can be considered specialised types of online recruitment firms where users signed up to make themselves available to participate in user research, usually for a fee. Researchers also sign up to recruit participants and conduct their research via the platform. Although it can be a very easy and quick way to not only recruit but also conduct the research, it may be difficult to find participants in a niche interest group who live in a particular geographic location.

d) Advertisements for new participant recruitment: An advert may be placed in any form of media or through any channel where the target participants are likely to be found, e.g., on a travel page if participants are frequent travellers. This also includes free or low-cost classified ads, social media ads etc. The advert must state that it is about the recruitment of

research participants, what the qualification criteria are, what the research entails, how long it will take, and what the incentive or payment will be. Do not mention what the product is, just focus on basic participant selection criteria.

e) Social media and online communities: Apart from paid social media ads and advertising on online community pages, requests to participate may also be posted on social media groups and online community pages, therefore social media marketing and online community development is not only important from a marketing perspective but also to recruit participants.

f) Special interest organisations, groups or events: Organisations, groups or events which represents a specific type of career, cause or field of interest, may be approached to identify relevant research participants, e.g. a teacher union or conference to test educational software, a fishing club to test a fishing app, or a music festival to test a site for a particular music genre.

g) Random selection at a public place: As a form of convenience sampling, potential participants could be approached at a public place where people who meet the selection criteria are likely to be present and asked if they will be willing to participate. Usually, the research will be conducted right there and then, therefore it is suitable for techniques such as interviews, surveys, card sorting, and other techniques which may be considered 'gorilla research'.

h) Database of existing users, customers, sales leads or staff of the firm or firm's partners: Existing customers of the firm or users of the product may be contacted and asked if they will be willing to participate. Another possibility is to ask the marketing or sales department for a list of sales leads, i.e. people who merely indicated an interest in the product in the recent past, which may be a very effective recruitment and sales strategy because existing customers will be bothered less, and it may be used as an opportunity to convince them to become customers. Sometimes it may be needed to dig deeper into the supply chain or corporate structures to ask for staff or clients of a supplier or buying firm of the firm you want to conduct research for.

i) Research participant panels: A panel of research participants is a database of usually existing users who indicated that they are willing to participate in research. They indicated their willingness through various means e.g., by opting in via a link or opt-in button on an email, the website, a landing page e.g. a competition, a request via social media, or any other recruitment process. They may also be previously recruited research participants who indicated that they are willing to participate in future research again. The database should contain basic information for each person, including name, contact detail, and some basic information to enable screening e.g., location, crucial demographics, characteristics that could link the person to a specific persona. Some research participants are particularly helpful in research, therefore an indication of whether a person was very helpful in previous research sessions may be very valuable, if the same participants are not used over and over again. Panel participants should not be used where it is important to test people who are not already familiar with the product or the brand.

i) Snowball recruitment: Ask a research participant if she can recommend a friend, colleague or family member who may also meet the selection criteria. This can be quick, easy, and costless, although an incentive may sometimes be offered to increase chances that the participant will give a proper thought into who in their social circle may qualify and take the 'social risk' of referring another person.

Screening and scheduling of participants

Although selection criteria were used to identify and approach participants, final screening is needed to make certain that the identified person indeed complies with the selection criteria, in other words, to finally 'qualify' them as a suitable research participant. It can be as simple as asking a few quick questions face-to-face in case participants are selected at a public place for immediate testing, or more often it entails a telephonic questioning or rather discussion around the participant selection criteria and whether the person meets those. Avoid talking about the product since the aim is only on getting the right person into the research session unless the product may raise moral objections, or it is of relevance in another way that the participant must know what it is about. Those who do not qualify may be entered into the research participant panel in case they agree and if it is likely that they may meet selection criteria for a future research assignment.

Then, after a person has qualified, mention what incentives are offered to enable the person to decide whether she will be interested to take this further.

Then, if the person is interested to participate, arrange a time and venue still during this same conversation. Schedule the research session at a time and location convenient to the participants, which may be outside normal working hours and not at the office of the researcher. Convenience for the participants should be more important than convenience for the researchers. Tools for scheduling may include Calendly which makes appointment scheduling very easy.

Incentives to participate

Ideally, participants should be compensated to cover their time, effort, and travel expense. Higher qualified, high-demand professionals may therefore be very expensive to recruit. However, often people may be happy with just a token of appreciation such as a gift hamper, chocolates, cold drink or a coffee, and travel expense if they had to travel to the research venue.

User testing

User testing as the most common UX research method

A User Test is a UX research technique where you give a prototype or a real app, website, software, or other digital product, to a person to use while being observed by you as the UX researcher or designer. You observe how the user interacts with the product, what they find easy or difficult to use or discover and ask them what they think of it and how it can be improved. The user who participates in the product test can be anyone who may represent the target user of the product. Note that it is the product that is tested, not the user. The product must be easy and valuable to use, so if the user struggles to use it or to find it valuable, we know that we must make amendments or completely redesign the product.

User tests can be moderated or unmoderated.

Moderated user tests are tests where you as the researcher moderate or lead the session in person with the user. You talk to the user directly. It is the most common UX research technique, and the remainder of this lesson will be devoted to this technique.

Unmoderated user testing is an automated test where software gives instructions to the user on how to use the product and measures the user's interactions automatically. Unmoderated user testing is a more advanced technique that will be discussed in a future lesson.

Venue for user testing

You may wonder where a user test should be performed. Ideally, it should be performed in the natural setting or place when and where a typical user would normally use the product. For example, if you test a gym app, you should test it in a gym where the user is exercising. However, this is a more advanced but probably the best UX research technique called participant observation or contextual enquiry, which will be discussed in a future lesson.

For now, we just focus on a basic user test that can be performed anywhere where you and the user feel comfortable. There is no urgency to use a 'usability lab'. Often better results are obtained in less formal settings e.g., a meeting room where researchers and the participant interact face-to-face, and where it is easy for project team members to simply pop in and share in the learning experience.

Tasks that users must perform during user testing

During a user test, the researcher or moderator provides the participant with several tasks to perform on the app or prototype that are being tested. Tasks may be to perform specific actions, for example, to register, create a profile or set some fitness goals on a gym app.

Tasks may also be to find items or content. For example, you may want the user of a gym app to search for a particular type of exercise or to get an answer to a frequently asked question.

A user test can take a long time if there are many tasks to be performed. Users may quickly get tired or bored, especially if they are not paid or otherwise incentivised to participate in the user test. Therefore, it is important to prioritise the tasks you would like the user to perform, putting the most important actions or finding the most important content or items first. Start with the most important tasks or features you want to test for, then move on to the less important ones. This is important to make certain there is sufficient time to cover the important tasks and to test them while the user is still motivated.

Start with general tasks so that it is possible to see if the user intuitively can orient themselves in the broader environment while they are still unfamiliar with the product. Then gradually move to increasingly more specific tasks to test very specific task flows or very specific features. For example, first, ask the user to generate a complete gym program, and only thereafter ask to perform sub-tasks of the greater task, like changing the resistance level of a particular piece of gym equipment.

Follow a logical order in which tasks are normally performed, although it may sometimes be useful to test coping with unexpected interruptions and task requests in an illogical order, especially in the order of common sequence interruptions in real life.

Start with simple tasks that are easy to perform so that the participant can build some confidence, then move to more complex tasks.

Tasks must fit into the natural context and frame from a realistic perspective. Avoid being abstract or unnatural.

Use more generic task descriptions. Avoid being too direct, to-the-point or specific to the product, or using exact words and navigation as found on the product interface. For example, tell a user ‘You want to exclude exercises that may place strain on your back, show me how you would accomplish this, rather than ‘go to your profile, click on exercise preferences, and disable back straining exercises’.

Similarly, avoid specific words used by the app, because it may provide too many clues on how to perform the task. For example, if you want to test if the user will be able to find a good eating plan to go with the gym program, don’t ask ‘How would you manage nutrition for your chosen gym plan?’ if it falls under the ‘manage nutrition tab’. Rather ask it in general terms not specific to the apps’ chosen words, for example, ‘How would you know what to eat?’.

However, sometimes it may be useful to be very specific, for example, when you zoom into very specific parts of tasks.

Try to make the tasks personally relevant and interesting, possibly in a short story format to make it clear.

Moderation of the user test

Moderating or guiding the user through a user test is a systematic process that includes the following steps.

Step 1: Briefing of the research team: Discuss the research purpose, methods, responsibilities, and rules for interaction before the participant arrives. The interacting researcher or moderator provide tasks to the user to do, while observers (which include observing researchers and other interested team members) are there just to observe and take notes, not to talk with the user or amongst themselves, although they may ask questions to the user after the user has completed all tasks. However, the interacting/moderating researcher may and should ask probing questions. Also, in case a usability study is combined with an interview or used for co-creation, i.e., where the focus is less on usability and more on co-creation and gaining rich insights, it is perfectly fine to discuss and ask questions.

Step 2: Welcome and brief the participant: Greet and make the participant feel welcome, briefly introduce the research team present and user to each other, brief the user about the testing procedure and roles of the researcher and the participant, mention you will ask them to perform tasks, but evaluation will be of the product’s ease of use and not about evaluating them. Ask the user to think out loud while performing the tasks, mention that it’s fine to quit any task or the entire session, don’t mention too much about the goals of the research and let the user sign a consent form. Make certain the participant is the right person who has been qualified for this specific research assignment.

Step 3: Moderating during the test session: Good moderation or conversation management through the main body of the user test session is important:

- Make the user feel comfortable and relaxed throughout the session.
- Give the user tasks to do, as discussed earlier.
- Ask the user to think out loud as they perform their tasks, by saying what they search for, what they are thinking of, how they reason, what they try to accomplish, what they expect to see or get, what they think the different terms or functionalities mean or will provide, what they are uncertain about, and what they like or dislike. Users tend to forget to think out loud, therefore, may need a constant reminder to speak up and probing on what they are thinking.
- Ask probing questions to uncover the users' thoughts, especially where they hesitate, look frustrated, uncertain, surprised, or delighted, or if they don't think out loud in sufficient detail. Typical probing questions overlap with what you ask users to think out loud about.
- Carefully observe what the user does, how tasks are approached, where you expect them to go and what they search for. Make notes of your observations during the session otherwise you may forget some important observations. Ideally, there should be one researcher moderating the test, and another researcher taking notes so that one person can focus only on fluent, well-directed moderation, and another person focuses on complete and clear notetaking. Important observations can be spoken out loud by the researcher if the session is recorded on video or especially on audio so that it gets the attention it deserves when the recordings are listened to later, or if other researchers might have missed those observations. Beware of distracting the participant, however.
- In case a user cannot perform a key step in a task, holding the user back from performing the rest of the task, it is fine to help the user so that usability of the rest of the task can be assessed, however, it should be made note of because identification of difficult-to-perform tasks is a key objective of user testing. Also pay close attention to what the thought processes are and what went wrong, and how it can be avoided. It may be useful to ask a lot of questions when a user can't perform a task rather than just giving away the answer to get unstuck. However, in case the reason for the inability to continue is due to a known issue or error that will be fixed, even if it has just been discovered, it is fine to simply help the user out of that situation.
- For particularly important tasks you may want to ask a user to rate how easy it was to perform that task, or how valuable that feature will be to them if it will be expensive or time-consuming to build or improve that feature.

Step 4: Post-test questions: After tasks have been completed, ask what the participant liked and disliked about the product, and how it can be improved. A Net Promoter Score question may also be asked, namely 'On a scale of 0 to 10, where 0 is not at all and 10 is very likely, how likely are you to recommend this product to a friend, and why?'. An important goal of the study is to assess the financial viability of the product, or if it is important to measure improvements over time. Recall can also be tested, for example, ask if the participant still remembers the content, where to look, what something is called or how to perform a certain task they performed a little earlier in the user test.

Step 5: Debrief the participant: After the user test has been completed, thank the user for their time and insights. You may then ask if the user would like to participate in possible future research, and if so, enter the user's details into a database or research participant panel. The incentive or payment for the user's time to participate in the user test are usually provided after the test session, however, it can also be given at the end or beginning of the session.

Step 6: Debrief the research team, and immediately or as soon as possible **capture** the insights gained while it is still fresh in memory, otherwise valuable insights may soon be forgotten. This is especially true if another user test will be performed soon because trying to remember observations after more than one session is very difficult and the researcher may get confused between the different test sessions.

Considering the principle of match between system and the real world, I would like you to take your book and create 3 different icons for elementary, secondary, and high school. If you had to put these institutions into an icon or symbol, what would they look like?

These icons should represent each of these institutions. Write down the reason on why you chose these specific icons and test these icons with friends and family to determine if your users will be able to recognise and match these icons to the real world.

You can either draw these icons in your sketch book or digital – which ever means you are comfortable with. Share your design on your favourite social media platform using the #Uxicondesign and provide some reasoning for your design choice.

Conclusion

In this lesson, we learned that UX is crucial for the success of a digital product because it helps to align the product with real user needs and ensure usability. We learned techniques for test user recruitment which is not an easy task. Lastly, we learned about user testing as the most common UX research technique and how to conduct a successful usability test yourself.

In the next lesson, we will learn about what to design, the business strategies behind successful digital products. We will learn about designing effective e-commerce sites, a one-sided digital marketplace. However, the digital economy is evolving fast towards more efficient, complex structures, therefore we will shift attention to two multi-sided platforms, ecosystems and digital business models, and how to design pleasant user experiences for all digital platform participants.

CEO of Iconfinder Martin Leblanc stated that a user interface is like a joke. If you have to explain it, it's not that good.

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